Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0209528362014022	1.0044167521097878 – 1.028552992418132

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9999389678238367	0.9988822449399083 – 1.0019213770644335

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.999545442181394	0.9989342808541711 – 1.0014338340055817

Thin QR

512 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0013490003755645	0.9992072846344657 – 1.0035722503949913

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9999680183567204	0.998100644669221 – 1.006039028938504

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9932030905284503	0.9862801868789893 – 1.007483188743173

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9959944736588449	0.9920063778899814 – 0.9988831974914898

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9947204933445353	0.9924569352137417 – 1.0028312037757108

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0003095017022594	0.9996913580246913 – 1.0024752475247525

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.002187844645146	0.9990965717274499 – 1.0042834705630383

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9988776130571014	0.9974802855676357 – 1.0012975171833356

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0000733170524125	0.9988657096941796 – 1.0012095566320618

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0003351722513611	0.9982475085060689 – 1.0032564327396971

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0022779043280183	1.0002841716396704 – 1.0113895216400912

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0021819561711414	0.992643247573273 – 1.0048934706273134

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9977473271394556	0.9967462797966133 – 1.0013204173921173

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0020833333333337	0.9979209979209976 – 1.004428236519927

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0033444816053512	1.000418235048097 – 1.0033444816053512

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0025886337406835	0.9989399875376913 – 1.0068572085058107

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9943369422082682	0.9896212753455892 – 1.001343488214958

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9991390368849621	0.994341539844011 – 1.0006276015852136

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9968366943455912	0.9819406051349345 – 1.0067972337839728

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9989854787615325	0.9964796177870741 – 1.008395017412424

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9975581228325775	0.9960705660840735 – 1.000474285478616